

DRP Journal

Aditya Pahuja

May 10, 2026

Contents

1	Rudiments of homotopy theory	2
2	Differential forms	3
3	Homology and cohomology	4
3.1	Homological algebra and the universal coefficients theorems	4
3.2	Products	4

1 Rudiments of homotopy theory

The n th homotopy group $\pi_n(X, x_0)$ is the set of homotopy classes of maps $(I^n, \partial I^n) \rightarrow (X, x_0)$, where I is the interval $[0, 1]$. Note that such maps correspond uniquely to maps $(S^n, s_0) \rightarrow (X, x_0)$ because $I^n/\partial I^n$ is homeomorphic to S^n . This set of homotopy classes is made into a group as follows: for maps $f, g: (I^n, \partial I^n) \rightarrow (X, x_0)$,

$$(f + g)(s_1, s_2, \dots, s_n) = \begin{cases} f(2s_1, s_2, \dots, s_n), & s_1 \in [0, \frac{1}{2}] \\ g(2s_1 - 1, s_2, \dots, s_n), & s_1 \in [\frac{1}{2}, 1]. \end{cases}$$

The additive notation suggests that this group is abelian, which is indeed true for $n > 1$:

Proposition 1.1. For $n \geq 2$, $\pi_n(X, x_0)$ is abelian.

Additionally, π_n is a functor: a map $\varphi: (X, x_0) \rightarrow (Y, y_0)$ induces the homomorphism

$$\varphi_*: \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0), \quad \varphi_*[f] = [\varphi \circ f].$$

The higher homotopy groups satisfy some nice elementary properties similar to those satisfied by the fundamental group:

Proposition 1.2. For all $n \geq 2$, the following properties hold.

- (a) If $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ is a covering map, $p_*: \pi_n(\tilde{X}, \tilde{x}_0) \rightarrow \pi_n(X, x_0)$ is an isomorphism.
- (b) If $(X_\alpha)_{\alpha \in A}$ is a set of topological spaces, there is an isomorphism $\pi_n(\prod_\alpha X_\alpha) \cong \prod_\alpha \pi_n(X_\alpha)$.

If X is path connected, and γ is a path from x_1 to x_0 , there is a change of basepoint isomorphism $\beta_\gamma: \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$. If we only consider loops γ based at x_0 , then the map sending $[\gamma] \in \pi_1(X, x_0)$ to $\beta_\gamma \in \text{Aut } \pi_n(X, x_0)$ is an action of π_1 on π_n . For $n \geq 2$ this action makes π_n into a module over the group ring $\mathbb{Z}[\pi_1]$.

Definition 1.3 (Relative homotopy groups) — Let $x_0 \in A \subseteq X$ for spaces A, X . The *relative homotopy group* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps $(D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0)$ under homotopies relative to ∂S^{n-1} , or, equivalently, maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$ under homotopies relative to ∂I^n , where J^{n-1} is the closure of the complement of a single face of I^n in ∂I^n .

Maps in the second sense can just as well be thought of as maps $f: (I^n, \partial I^n) \rightarrow (X, A)$ such that $f(s_1, \dots, s_{n-1}, 0) = x_0$; thus $\pi_n(X, A, x_0)$ can be made into a group for $n \geq 2$ (which is abelian for $n \geq 3$) by the usual composition operation. Like with homology, these relative homotopy groups fit into a long exact sequence.

Proposition 1.4. Let (X, A, B) be a triple of spaces (so $B \subseteq A \subseteq X$) and $x_0 \in B$. There is an exact sequence

$$\cdots \rightarrow \pi_{n+1}(X, A, x_0) \xrightarrow{\partial} \pi_n(A, B, x_0) \xrightarrow{i_*} \pi_n(X, B, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, B, x_0) \rightarrow \cdots$$

where $i: (A, B, x_0) \hookrightarrow (X, B, x_0)$ and $j: (X, B, x_0, x_0) \hookrightarrow (X, A, x_0)$ are the inclusions and the connecting homomorphism ∂ is induced by the restriction of a map $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$ to a map $(I^{n-1}, \partial I^{n-1}) \rightarrow (A, x_0)$, with I^{n-1} embedded in I^n as the face that was deleted to make J^{n-1} .

In the case where $B = x_0$ this gives an exact sequence

$$\cdots \rightarrow \pi_{n+1}(X, A, x_0) \xrightarrow{\partial} \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \rightarrow \cdots$$

Proof. First we check exactness at $\pi_n(A, B, x_0)$. The composition $i_*\partial = 0$, since given continuous map $f: (I^{n+1}, \partial I^{n+1}, J^n) \rightarrow (X, A, x_0)$, the restriction $f|_{I^n}: (I^n, \partial I^n) \rightarrow (A, B)$ is nullhomotopic relative to ∂I^n via f itself because $f(J^n) = x_0$. To check that $\ker i_* = \text{im } \partial$, suppose that $f: (I^n, \partial I^n, J^{n-1}) \rightarrow (A, B, x_0)$ is such that $i \circ f: (I^n, \partial I^n, J^{n-1}) \rightarrow (X, B, x_0)$ is nullhomotopic relative to ∂I^n . Then there is a homotopy of $i \circ f$ to the constant map equalling x_0 , which yields a continuous map out of I^{n+1} such that ∂I^{n+1} maps into B and J^n $\square$.

Next we check exactness at $\pi_n(X, B, x_0)$. $\square$

2 Differential forms

Definition 2.1 (Differential form) — Let M be an n -manifold and $\Lambda^k T^*M$ the k th exterior power of the cotangent bundle T^*M . A *differential k -form* (often just called a *k -form*) is a section ω of $\Lambda^k T^*M$. The vector space of k -forms on M is denoted $\Omega^k(M)$, and the graded algebra of all differential forms on M is denoted $\Omega^*(M)$.

In local coordinates $(x_1, \dots, x_n)$, a k -form ω is expressible as

$$\sum_{1 \leq i_1 < \dots < i_k \leq n} A_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

where the $A_{i_1, \dots, i_k}: U \rightarrow \mathbb{R}$ are smooth functions defined in the coordinate neighborhood.

The multiplication operation in $\Omega^*(M)$ is given by just wedging the forms together; in local coordinates, this looks like

$$(f dx_{i_1} \wedge \dots \wedge dx_{i_k}) \wedge (g dx_{j_1} \wedge \dots \wedge dx_{j_\ell}) = fg dx_{i_1} \wedge \dots \wedge dx_{i_k} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_\ell}$$

and extend by linearity for more complicated-looking forms.

The algebra of k -forms comes with an *exterior differentiation* operator, which makes it into a cochain complex known as the *de Rham complex*.

Proposition 2.2. There exists a unique endomorphism $d: \Omega^*(M) \rightarrow \Omega^*(M)$ sending k -forms to $(k+1)$ -forms such that

- (i) $d^2 = 0$;
- (ii) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^{\deg \omega} \omega \wedge d\eta$;
- (iii) if $f \in C^\infty(M) = \Omega^0(M)$, then $df: TM \rightarrow \mathbb{R}$ is the differential of f .

Proof. Over open subsets of $\mathbb{R}^n$, we can define

$$d(f dx_{i_1} \wedge \dots \wedge dx_{i_k}) = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j \wedge dx_{i_1} \wedge \dots \wedge dx_{i_k}$$

and extend by linearity; this is easily seen to satisfy the necessary properties. For an arbitrary n -manifolds, take a partition of unity subordinate to a covering of the manifold by coordinate neighborhoods; the properties in the proposition are well-behaved under linear combination, so summing over the partition of unity lets us define d on any manifold.

Now let d and $\tilde{d}$ be two endomorphisms of $\Omega^*(M)$ with the given properties $\square$

3 Homology and cohomology

A *simplex* of dimension n is the simplest n -dimensional polytope, analogous to how the triangle is the simplest polygon. The canonical choice of simplex is

$$\Delta^n := \{(t_0, \dots, t_n) : 0 \leq t_i, t_0 + \dots + t_n \leq 1\}.$$

the convex hull of the endpoints of the n coordinate vectors.

3.1 Homological algebra and the universal coefficients theorems

The goal is to algebraically relate integral homology, homology with coefficients in a particular group G , and cohomology with values in G . This manifests as relating the homology of the singular chain complex

$$\cdots \rightarrow C_{n-1}(X) \rightarrow C_n(X) \rightarrow C_{n+1}(X) \rightarrow \cdots$$

to that of the complexes

$$\begin{aligned} \cdots \rightarrow C_{n+1}(X) \otimes G \rightarrow C_n(X) \otimes G \rightarrow C_{n-1}(X) \otimes G \rightarrow \cdots \\ \cdots \rightarrow \text{Hom}(C_{n-1}(X), G) \rightarrow \text{Hom}(C_n(X), G) \rightarrow \text{Hom}(C_{n+1}(X), G) \rightarrow \cdots \end{aligned}$$

(where $\otimes$ and Hom are over $\mathbb{Z}$).

Proposition 3.1. The functor $- \otimes G$ is right-exact; the functors $\text{Hom}(-, G)$ and $\text{Hom}(G, -)$ are left-exact.

Proof. Let $0 \rightarrow A \xrightarrow{u} B \xrightarrow{v} C \rightarrow 0$ be a short exact sequence of abelian groups. We can see that $v \otimes 1: B \otimes G \rightarrow C \otimes G$ is surjective because given $c \in C$ and $b \in v^{-1}(c)$, $(v \otimes 1)(b \otimes g) = c \otimes g$. To get exactness at $B \otimes G$, we note that

$$((v \otimes 1) \circ (u \otimes 1))(a \otimes g) = vu(a) \otimes g = 0$$

for all simple tensors $a \otimes g \in A \otimes G$, so $\ker(v \otimes 1)$ contains $\text{im}(u \otimes 1)$. In the other direction, suppose that $\sum_i b_i \otimes g_i \in \ker(v \otimes 1)$. If $b_i = u(a_i)$, then $\sum_i b_i \otimes g_i = u \otimes 1(\sum_i a_i \otimes g_i)$, hence $\sum_i b_i \otimes g_i \in \text{im}(u \otimes 1)$; thus $- \otimes G$ is right-exact.

The proof for Hom is similar. □

The main theorem is the following:

Theorem 3.2 (Universal coefficients formulas). Let X be a topological space, n a nonnegative integer, and G an abelian group. Then the sequences

$$\begin{aligned} 0 \rightarrow H_n(X) \otimes G \rightarrow H_n(X; G) \rightarrow \text{Tor}_{\mathbb{Z}}(H_{n-1}(X), G) \rightarrow 0, \\ 0 \rightarrow H^n(X; \mathbb{Z}) \otimes G \rightarrow H^n(X; G) \rightarrow \text{Tor}_{\mathbb{Z}}(H^{n+1}(X; \mathbb{Z}), G) \rightarrow 0, \\ 0 \rightarrow \text{Ext}(H_{n-1}(X), G) \rightarrow H^n(X; G) \rightarrow \text{Hom}(H_n(X), G) \rightarrow 0 \end{aligned}$$

are split exact.

3.2 Products